

HOMEWORK 9 SOLUTION

Problem 1

Consider the Bernoulli regression model,

$$P(y_i = 1) = p_i = \frac{e^{x_i\beta}}{1 + e^{x_i\beta}}, \quad y_i \in \{0, 1\}, \quad i = 1, \dots, n,$$

with β a one dimensional unknown parameter.

- (1) (10 pts) Write down the log likelihood function $L(\beta)$ for β .

Solution:

$$\begin{aligned} P(y|\beta) &= \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i} \\ L(\beta) &= \log P(y|\beta) = \sum_{i=1}^n (y_i \log p_i + (1 - y_i) \log(1 - p_i)) \\ &= \sum_{i=1}^n [y_i(x_i\beta - \log(1 + e^{x_i\beta})) + (1 - y_i)(-\log(1 + e^{x_i\beta}))] \\ &= \beta \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \log(1 + e^{x_i\beta}) \end{aligned}$$

- (2) (10 pts) Find both $dL/d\beta$ and $d^2L/d\beta^2$.

Solution:

$$\begin{aligned} L'(\beta) &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \frac{x_i e^{x_i\beta}}{1 + e^{x_i\beta}} = \sum_{i=1}^n x_i (y_i - p_i) \\ L''(\beta) &= - \sum_{i=1}^n \frac{x_i^2 e^{x_i\beta} (1 + e^{x_i\beta}) - (x_i e^{x_i\beta})^2}{(1 + e^{x_i\beta})^2} = - \sum_{i=1}^n \frac{x_i^2 [e^{x_i\beta} + (e^{x_i\beta})^2 - (e^{x_i\beta})^2]}{(1 + e^{x_i\beta})^2} \\ &= - \sum_{i=1}^n x_i^2 \frac{e^{x_i\beta}}{1 + e^{x_i\beta}} \frac{1}{1 + e^{x_i\beta}} = - \sum_{i=1}^n x_i^2 p_i (1 - p_i) \end{aligned}$$

- (3) (10 pts) Write down a Newton–Raphson algorithm for finding the maximum likelihood estimator $\hat{\beta}$.

Solution: With a chosen starting point $\beta^{(1)}$, perform iteratively

$$\beta^{(t+1)} = \beta^{(t)} - \frac{L'(\beta^{(t)})}{L''(\beta^{(t)})} = \beta^{(t)} + \frac{\sum_{i=1}^n x_i (y_i - p_i^{(t)})}{\sum_{i=1}^n x_i^2 p_i^{(t)} (1 - p_i^{(t)})}$$

where

$$p_i^{(t)} = \frac{e^{x_i\beta^{(t)}}}{1 + e^{x_i\beta^{(t)}}}.$$

- (4) (10 pts) If $\hat{p}_i = \exp(x_i \hat{\beta}) / (1 + \exp(x_i \hat{\beta}))$, show how to obtain the approximation $E(\hat{\beta}) = \beta$ and

$$\text{Var}(\hat{\beta}) = \frac{1}{\sum_{i=1}^n x_i^2 \hat{p}_i (1 - \hat{p}_i)}.$$

Solution:

$\hat{\beta}$ minimizes $L(\beta)$, so $L'(\hat{\beta}) = 0$. Use the first order Taylor expansion approximation for $L'(\hat{\beta})$,

$$0 = L'(\hat{\beta}) \approx L'(\beta) + (\hat{\beta} - \beta)L''(\beta) \quad \Leftrightarrow \quad \hat{\beta} \approx \beta - \frac{L'(\beta)}{L''(\beta)}$$

$$E(L'(\beta)) = \sum_{i=1}^n x_i E(y_i - p_i) = 0 \quad \Rightarrow \quad E(\hat{\beta}) \approx \beta - \frac{E(L'(\beta))}{L''(\beta)} = \beta$$

$$\begin{aligned} \text{Var}(\hat{\beta}) &\approx \text{Var}\left(\frac{L'(\beta)}{L''(\beta)}\right) \approx \frac{\text{Var}(L'(\beta))}{L''(\beta)^2} = \frac{\sum_{i=1}^n x_i^2 \text{Var}(y_i)}{(\sum_{i=1}^n (x_i^2 p_i (1 - p_i)))^2} \\ &= \frac{\sum_{i=1}^n x_i^2 p_i (1 - p_i)}{(\sum_{i=1}^n (x_i^2 p_i (1 - p_i)))^2} \approx \left(\sum_{i=1}^n x_i^2 \hat{p}_i (1 - \hat{p}_i)\right)^{-1} \end{aligned}$$

- (5) (10 pts) Hence, describe a test for the hypothesis $H : \beta = 0$.

Solution:

Under the null hypothesis $H : \beta = 0$, we have approximately

$$\hat{\beta} \sim N\left(0, \left(\sum_{i=1}^n x_i^2 \hat{p}_i (1 - \hat{p}_i)\right)^{-1}\right),$$

so the test statistic is approximately a standard normal variable;

$$z = \frac{\hat{\beta} - \beta}{\sqrt{\text{Var}(\hat{\beta})}} = \hat{\beta} \left(\sum_{i=1}^n x_i^2 \hat{p}_i (1 - \hat{p}_i)\right)^{\frac{1}{2}}$$

We can reject the null hypothesis if $|z| > z_{\alpha/2}$

Problem 2

The Bernoulli regression model, as given in Problem 1, is analyzed using a Bayesian approach and the prior for β , i.e. $\pi(\beta)$, is chosen to be normal with mean 0 and variance σ^2 .

- (1) (10 pts) Write down the posterior density (up to a constant of proportionality) for β in terms of the (x_i, y_i) .

Solution:

$$\begin{aligned} \pi(\beta \mid \text{data}) &= \frac{P(y_{1:n} \mid \beta) \pi(\beta)}{P(\text{data})} \propto \prod_{i=1}^n P(y_i \mid \beta) \pi(\beta) \\ &= \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1 - y_i} N(\beta \mid 0, \sigma^2) \\ &\quad \text{with} \quad p_i = \frac{e^{x_i \beta}}{1 + e^{x_i \beta}} \end{aligned}$$

Or equivalently,

$$\pi(\beta \mid \text{data}) \propto \exp\left(-\frac{1}{2}\beta^2/\sigma^2\right) \prod_{i=1}^n \frac{e^{y_i x_i \beta}}{1 + e^{x_i \beta}}$$

- (2) (10 pts) A way to sample from a density directly is available if the logarithm of the density is concave. Show that the log of the posterior density from part (1) is concave.

Solution:

$$\begin{aligned} l(\beta) &= \log(\pi(\beta \mid \text{data})) = \log(P(y_{1:n} \mid \beta)) + \log(\pi(\beta)) - \log(P(\text{data})) \\ &= L(\beta) + \log(N(\beta \mid 0, \sigma^2)) - c = L(\beta) - \log(\sqrt{2\pi}) - \frac{\beta^2}{2\sigma^2} - c \end{aligned}$$

$$l'(\beta) = L'(\beta) - \frac{\beta}{\sigma^2} = \sum_{i=1}^n x_i(y_i - p_i) - \frac{\beta}{\sigma^2}$$

$$l''(\beta) = L''(\beta) - \frac{1}{\sigma^2} = -\sum_{i=1}^n x_i^2 p_i(1 - p_i) - \frac{1}{\sigma^2} < 0$$

So $l(\beta)$ is concave.

- (3) (10 pts) If the prior is taken to be improper, i.e. $\pi(\beta) \propto 1$, which would follow if σ is taken to infinity, what is the relationship between the posterior and the maximum likelihood estimator.

Solution:

$$\pi(\beta \mid \text{data}) = \frac{P(y_{1:n} \mid \beta)\pi(\beta)}{P(\text{data})} \propto P(y_{1:n} \mid \beta)$$

The posterior density of β will be proportional to the likelihood function when $\pi(\beta) \propto 1$, so maximizing $\pi(\beta \mid \text{data})$ is equivalent to maximizing $L(\beta)$.

Therefore, the mode of the posterior will be the MLE estimator $\hat{\beta}$, which maximizes $L(\beta)$.

- (4) (10 pts) With this improper prior, the aim is to derive the Bayes factor for the choice of null hypothesis $H_0 : \beta = 0$ versus the alternative $\beta = 1$. Show that the log Bayes factor can be written as

$$\log B = \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \log\left(\frac{1 + e^{x_i}}{2}\right).$$

Solution:

$$\begin{aligned} \log B &= \log\left(\frac{m(y \mid \beta = 1)}{m(y \mid \beta = 0)}\right) = L(\beta = 1) - L(\beta = 0) \\ &= \sum_{i=1}^n x_i y_i - \log(1 + e^{x_i}) + \sum_{i=1}^n \log(1 + e^0) \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \log\left(\frac{1 + e^{x_i}}{2}\right) \end{aligned}$$

- (5) (10 pts) Find the approximate normal distribution of $\log B$ under the null hypothesis.

Solution:

$y_i \sim \text{Bernoulli}(0.5)$ under the null hypothesis, so

$$E(\log B) = \frac{1}{2} \sum_{i=1}^n x_i - \sum_{i=1}^n \log\left(\frac{1 + e^{x_i}}{2}\right), \quad \text{Var}(\log B) = \frac{1}{4} \sum_{i=1}^n x_i^2$$

so approximately

$$\log B \sim N \left(\frac{1}{2} \sum_{i=1}^n x_i - \sum_{i=1}^n \log \left(\frac{1 + e^{x_i}}{2} \right), \quad \frac{1}{4} \sum_{i=1}^n x_i^2 \right)$$

Since the central limit theorem applies.